

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Introduction to internet of things [NPT_4129]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A smart agriculture system uses IoT sensors for soil moisture monitoring. Farmers report that data updates are often delayed or missing. Identify the possible connectivity issues and suggest improvements. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Analyse the differences between MQTT and CoAP in IoT communication. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Discuss the home automation system using Arduino, where a temperature sensor controls a DC fan as an actuator (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply the caching workflow in a Sensor-Cloud architecture. Where should cache be placed, and what replacement strategies are commonly used? (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A self-driving car uses multiple cameras mounted at different angles as part of its IoT ecosystem. Apply your understanding to tell how IoT enables the communication and coordination between these sensors and the cloud platform. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analysing the concepts of fog computing and smart city concepts, explain if fog computing complements cloud computing in Smart City IoT deployments. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyze the concept of connected vehicles to explain how modern cars can prevent rear-end collisions in dense traffic conditions. (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Apply IIoT system requirements to recommend the key hardware and software components needed for setting up an intelligent production line. (5)
- Course Outcome: "CO4" / Cognitive Level: "Application"

5) Analyse IoT data analytics to predict crop health or pest infection from environmental parameters. (5)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Apply IoT data analytics to derive insights about user health patterns from wearable sensor data. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4129:Introduction to internet of things

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4129:Introduction to internet of things

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)